



HTML AND CSS

Detailed Roadmap

Training Plan Overview

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Meet Your Instructor





- Name: Ali Gamal
- Graduated: 2017, Faculty of Computer Science & Artificial Intelligence — Benha University
- Experience: 7+ Years in Web Development

MEET YOUR INSTRUCTOR

Professional Experience



MEET YOUR INSTRUCTOR

Training & Instructional Experience

1

FEB 2025 – PRESENT

Web Instructor at **STEM Entrepreneurship Center**

2

AUG 2023 – PRESENT

Web Instructor at **ITI (Information Technology Institute)**

3

JUL 2020 – JUL 2021

Web Instructor at **Courzerve**

Course Objectives



WHY?

We believe everyone should
have the power to bring their
ideas to life online — and it all
starts with HTML & CSS.



OBJECTIVES

What You Will Be Able
to Build and Do by the
End of This Course :



COURSE OBJECTIVES

Understand how the web works (HTML structure & CSS styling)

Build complete static websites using HTML5 & CSS3

Use semantic HTML elements for accessibility and SEO

Apply custom styles using fonts, colors, spacing, and layout

Build responsive web layouts using Flexbox and media queries

Create real-world projects from scratch

Who is This
Course For?



This course is designed for

Complete beginners with no prior coding experience.

Students studying computer science, media, or any tech-related field.

Designers who want to understand how to turn designs into real web pages

Anyone looking to switch careers and step into the world of web development



Tools You'll Need



TOOLS YOU'LL NEED

1. Visual Studio Code

Free and powerful code editor

We'll use it to write and organize our HTML & CSS

2. Google Chrome (or any modern browser)

For previewing your web pages

Includes built-in developer tools

3. Live Server Extension (Optional)

Instantly see your changes in the browser

Helps you develop faster

4. CodePen / GitHub (Optional)

Online editors and portfolio tools

You can use them to share your projects later

Don't worry if you've never
used these before

we'll walk through
everything step
by step.



Course Schedule



Training Plan

SESSION 1

HTML Basics + Intro to CSS

SESSION 2

HTML5 Semantic Tags + CSS Essentials

SESSION 3

CSS3 Styling, Flexbox,
Responsive Design

COURSE SCHEDULE

Notes :

Each session is 6 hours: Half for teaching, half for hands-on tasks

Practical tasks include building real pages (CV, landing page, one-page site)

The final session wraps up with a full mini project



You won't just
learn — you'll build
something real in
every session.

Let's Pause Here
— Any Questions
So Far?



Feel free to ask —
no question is too small!

S01

HTML Basics + Intro to CSS



S01

Session Goals



SESSION GOALS

By the end of this session,
you will be able to

- Understand how websites are structured using HTML
- Write clean HTML pages using basic tags (headings, paragraphs, images, links...)
- Organize Code & content using divs
- Link HTML and CSS files properly
- Apply basic CSS to style elements (colors, fonts, spacing)
- Create your first web page structure from scratch

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Topics Covered



Topics Covered

These are the main topics we'll cover today to help you build your first web page

01. What is HTML?

- How the web works
- Brief History of HTML
- Structure of an HTML document
- Mini Task

02. Common HTML Elements

- Headings, paragraphs
- Block and Inline Elements
- HTML Attributes
- Images
- Comments
- links, lists, buttons
- div, span, tables (basic)
- Form Tags

03. CSS Introduction

- What is CSS?
- Inline vs Internal vs External styles
- Basic selectors & properties
- Applying colors, fonts, and spacing
- The concept of the Box Model

04. Linking HTML & CSS

- Creating a proper folder structure
- Linking CSS to HTML
- Live Server overview

05. Hands-On Tasks

- **Task 1** : Build Your First HTML
- **Task 2** : An image (Use one local and one online image)
- **Task 3** : Clickable Profile Image
- **Task 4**: Build a simple HTML page (CV or Profile Card)
- **Task 5**: Add headings, paragraphs, image, link, and list
- **Task 6**: Style it with colors, fonts, and spacing using CSS
- **Bonus**: Use div and class to group and style sections

S01

How the web works



WHAT IS HTML?

How the Web Works

When you visit a website, your browser sends a request to a server.

The server responds with HTML, CSS, and JS files.

Your browser reads these files and builds the page you see.

→ HTML is the structure (what to show)

→ CSS is the style (how it looks)

→ JS is the behavior (what it does)

WHAT IS HTML?

Think of a Web Page Like a Car

HTML is the **car's frame and body** — the structure everything is built on

CSS is the **paint, wheels, and interior design** — it makes the car look good

JavaScript is the **engine and electronics** — it makes the car move and respond

*"Just like a car needs a frame, a style, and movement — your webpage needs **HTML** for structure, **CSS** for styling, and **JavaScript** for interactivity."*

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Brief History of HTML



Brief History of HTML



HTML = HyperText Markup Language

HyperText = Text that contains links to other documents
(like web pages)

Markup Language = A system that uses **tags** to describe the structure and format of content

HTML **describes the content and format** of a web page using tags

A **tag** is like a label that tells the browser what a piece of content is (heading, paragraph, image, etc.)

Tag Types :

1. One-Sided Tag (Self-closing) :

Doesn't need an end tag.

Example: `<img />`, `<br />`, `<hr />`

2. Two-Sided Tag:

Has an opening and a closing tag.

Example: `<p>Hello</p>`, `<h1>Title</h1>`

Is HTML enough to build a website?

✗ **No.** HTML alone is not enough.

It only gives structure, but a website also needs:

- CSS to make it look good (colors, layout, fonts...)
- JavaScript to make it interactive (buttons, sliders, forms...)

💡 *So HTML is the foundation, but not the full house.*

S01

Structure of an HTML document



STRUCTURE OF ANY WEB PAGE

Structure of Any Web Page

Any web page must follow a specific structure so the browser can read and render it correctly.

Just like a house needs a foundation, walls, and rooms — a web page also has a basic layout it must follow.

STRUCTURE OF ANY WEB PAGE

Structure of Any Web Page

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>My First Page</title>
```

```
  </head>
```

```
  <body>
```

```
    <h1>Hello, World!</h1>
```

```
    <p>This is my first web page.</p>
```

```
  </body>
```

```
</html>
```

<!DOCTYPE html> → Declares the document type (HTML5)

└─── <html> → The root element for everything in the page

└─── <head> → Contains meta information, title, etc..

└─── <body> → Contains all visible content (text, images, etc.)

S01

Mini Task



MINI TASK : LET'S TRY IT NOW! — YOU HAVE 10 MINUTES. (RAISE YOUR HAND IF YOU NEED HELP.)

Mini Task: Build Your First HTML Page

Objective:

Create a basic HTML file with the correct structure and some simple content.

What to Do:

1. *Open **VS Code***
2. *Create a file named **index.html***
3. *Add the standard **HTML structure***
4. *Inside **<head>** : Add a **<title>** with your name*
5. *Inside **<body>**: Add an **<h1>** with a welcome message & Add a **<p>** with a short introduction*

Bonus (Optional):

*Try adding **<hr />** or **
** inside the page and see what happens*

SO1

Headings, paragraphs



HEADINGS, PARAGRAPHS

Headings:

Used to define titles and sections in your page Ranges from `<h1>` to `<h6>`, where:

`<h1>` is the largest and most important.

`<h6>` is the smallest and least important

Paragraphs:

The `<p>` tag is used to define a block of text

`<p>`This is a paragraph of text. It's used for longer content or explanations.`</p>`



Notice

Did You Notice This?

When we added a `<p>` tag under a `<h1>` tag... Did anyone notice something?

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Block vs Inline Elements



Understanding Block vs Inline Elements

Every HTML tag is either a **Block** Element or an **Inline** Element by default. This affects how elements are displayed on the page.

Block Elements:

- *Always start on a new line*
- *Take up the full width of their parent container*
- *Common examples: `<h1>` to `<h6>`, `<p>`*

Inline Elements:

- *Stay in the same line with other content*
- *Only take up as much width as needed*
- *Common examples: We haven't fully covered inline tags like `<a>`, `<span>`, or `<img>` yet.*
- *We'll explore them in detail later — for now, just focus on how they behave on the page.*

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HTML Attributes



What Are Attributes?

Attributes are extra pieces of information added to HTML tags.

Think of it like ordering coffee:

- *The coffee itself is the tag: <coffee>*
- *But you want to add extra info: size, sugar, milk, etc.*
- *So your full order becomes:*

`<coffee size="large" sugar="2" milk="yes" />`

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Images



IMAGES

What is `<img>`?

The `<img>` tag is used to embed an image in a web page.

It is a self-closing tag (doesn't wrap content inside it).

Syntax :

```

```

- **src** : The source of the image (URL or file path)
- **alt** : Alternative text (if image fails or for screen readers)
- ***width*** : Sets width
- ***height*** : Sets height
- ***title*** : Shows a tooltip on hover



Notice

Did You **Know?**

So far, we've been adding local images from our own devices...
But what if you want to use a picture from the internet without
downloading it?

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Homework Task



HOMEWORK TASK

Mini Task: Build Your First HTML Page

Objective:

Create a basic HTML file with the correct structure and some simple content.

What to Do:

1. *Open **VS Code***
2. *Create a file named **index.html***
3. *Add the standard **HTML structure***
4. *Add an image using an online image URL (not from your PC)*
5. *Make sure to use: (src, alt, title, width and height)*

Bonus (Optional):

Add two images: one from your device (local), and one from online.

HOMEWORK TASK

Mini Task: Build Your First HTML Page

Submission:

Submit your task before Tuesday, 01/07/2025.

Include 2 screenshots:

- *Your HTML code*
- *The result in the browser*

Important Submission Notes :

- *Your full name must be included with the task*
- *Make sure your screenshots are clear*

Reminder :

- *I'll be checking each submission individually and providing personal feedback*

Tip :

- *You can use Snipping Tool, Snip & Sketch, or press Print Screen (PrtScn) on your keyboard to capture screenshots easily.*

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HTML Comments



What Are Comments?

Comments let you add notes to your code without affecting how the page works.

They are ignored by the browser, but help you or others understand the code.

Can be on one line or spanning multiple lines

Syntax :

```
<!-- This is a comment -->
```

Pro Tip:

Note: Don't delete sections — use comments instead. Clients change their minds, and comments let you restore code quickly.

SO1

links



LINKS

What Are links?

The `<a>` tag stands for “anchor”, and it's used to create clickable links.

Syntax :

```
<a href="https://wikipedia.org" target="_blank">Learn more</a>
```

Tip:

- href: stands for “hyperlink reference” → tells the browser where to go
- target="_blank": opens the link in a new tab
- The text between the tags is what the user clicks

S01

Mini Task



MINI TASK : LET'S TRY IT NOW! — YOU HAVE 10 MINUTES. (RAISE YOUR HAND IF YOU NEED HELP.)

Mini Task: Build Your First HTML Page

Objective:

Clickable Profile Image

When the user clicks the image, it should take them to a specific website or social media profile.

Requirements :

1. *Open **VS Code***
2. *Create a file named **index.html***
3. *Add the standard **HTML structure***
4. *You need to combine two elements*
5. *The image must be "inside" the clickable part*

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lists



LISTS

What Are Lists?

Lists are used to group related content together in a structured way.

There are 3 main types in HTML:

1- Ordered List - `<ol>`

2- Unordered List - `<ul>`

3- Description List - `<dl>`

Syntax :

```
<ol>
```

```
  <li>HTML</li>
```

```
  <li>CSS</li>
```

```
  <li>JavaScript</li>
```

```
</ol>
```

Syntax :

```
<ul>
```

```
  <li>Facebook</li>
```

```
  <li>Instagram</li>
```

```
  <li>Twitter</li>
```

```
</ul>
```

Syntax :

```
<dl>
```

```
  <dt>HTML</dt>
```

```
  <dd>A markup language</dd>
```

```
  <dt>CSS</dt>
```

```
  <dd>Used for styling</dd>
```

```
</dl>
```

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Text Tags



TEXT TAGS

Text Formatting Tags in HTML

`<span>` : Useful for targeting part of a sentence

Syntax :

`<p>This is a <span class="highlight">highlighted</span> word.</p>`

`<strong>` : Makes text bold

Syntax :

`<p>Please <strong>read this carefully</strong> before submitting.</p>`

TEXT TAGS

Text Formatting Tags in HTML

`<em>` : Makes text italic

Syntax :

`<p>I <em>really</em> like HTML!</p>`

`<del>` : Shows deleted text with a line-through

Syntax :

`<p>Old Price: <del>$50</del> New Price: $30</p>`

TEXT TAGS

Text Formatting Tags in HTML

`<br>` : Inserts a line break (like pressing "Enter") inside text.

Syntax :

```
<p>This is line one.<br>This is line two.</p>
```

`<hr>` : Inserts a horizontal line (used to separate content)

Syntax :

```
<p>Section above</p>
```

```
<hr />
```

```
<p>Section below</p>
```

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Div Tag



DIV TAG

The `<div>` Tag — The Invisible Container

`<div>` tag stands for "division" : It's a block-level element used to group other elements together in a section.

Syntax :

```
<div>
```

```
  <h2>About Me</h2>
```

```
  <p>I'm a web developer based in Cairo.</p>
```

```
</div>
```


S01

Button



BUTTON

The `<button>` Tag — Interactive Element

The `<button>` tag is used to create a clickable button on a web page.

Syntax :

```
<button>Click Me</button>
```

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HTML Tables



What is a Table?

A table displays information in rows and columns, just like an Excel sheet.

Syntax :

```
<table>
```

```
  <tr>
```

```
    <th>Name</th>
```

```
    <th>Age</th>
```

```
  </tr>
```

```
  <tr>
```

```
    <td>Ali</td>
```

```
    <td>30</td>
```

```
  </tr>
```

```
</table>
```

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HTML Forms



HTML FORMS

What is a Form?

A form allows the user to send data to a server (like name, email, etc.)

Syntax :

```
<form action="/submit" method="post">
```

```
  <!-- form elements go here -->
```

```
</form>
```

What is a Form?

`<input>` : Used for text, email, number, etc.

Syntax :

```
<input type="text" placeholder="Enter your name" />
```

Types :

- *text* : Single-line input
- *email* : Validates email format
- *password* : Hides input characters
- *number* : Numeric input
- *checkbox* : Multiple choice
- *radio* : Single choice

HTML FORMS

What is a Form?

`<label>` : Used to describe the input and link to it using for.

Syntax :

```
<label for="username">Username:</label>
```

```
<input id="username" type="text" />
```

`<textarea>` : Multi-line input box.

Syntax :

```
<label for="message">Username:</label>
```

```
<textarea id="message" rows="4" cols="30"></textarea>
```


What is a Form?

<select> + <option> : Dropdown list

Syntax :

```
<select>
```

```
  <option>Egypt</option>
```

```
  <option>Saudi Arabia</option>
```

```
</select>
```

<button> : Used to submit the form or perform actions

Syntax :

```
<button type="submit">Send</button>
```

HTML FORMS

What is a Form?

Real Example :

```
<form>
```

```
  <label>Name:</label>
```

```
  <input type="text" />
```

```
  <label>Email:</label>
```

```
  <input type="email" />
```

```
  <label>Message:</label>
```

```
  <textarea></textarea>
```

```
  <button type="submit">Submit</button>
```

```
</form>
```

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Introduction to CSS





What You Need to Know Before Writing CSS

WHAT YOU NEED TO KNOW BEFORE WRITING CSS

What You Need to Know Before Writing CSS?

CSS is about Style, but Style needs Structure.

You Need to Know:

- 01. HTML Structure Matters : You must know which elements you want to style
- 02. Selectors are the Key : You can't style anything unless you know how to select it.

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What is CSS?



WHAT IS CSS?

What is CSS?

CSS stands for Cascading Style Sheets.

It's the language used to style and design HTML content.

What Can CSS Do?

- Change colors
- Adjust fonts & sizes
- Control spacing (margin/padding)
- Add borders & backgrounds
- Create layouts (columns, grids)
- Add hover effects and transitions
- Make websites responsive

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Sources of Styles



SOURCES OF STYLES

Sources of CSS Styles

CSS Styles Can Come From 3 Main Sources:

1. ***Inline CSS*** : Inside the HTML element using style
2. ***Internal CSS*** : Inside a `<style>` tag in `<head>`
3. ***External CSS*** : In a separate `.css` file

Pro Tip:

Always separate structure and style

Keep your HTML for content, and CSS in its own file.

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Optional Task



OPTIONAL TASK:

Try writing the same style (color: purple) in:

1. *Inline*
2. *Internal <style>*
3. *External file*

Watch which one takes priority!

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Selectors



CSS Selectors — How to Target Elements

A selector tells the browser which element(s) to apply the CSS styles to.

- 1. **Tag selector (*p*)** : All `<p>` elements
- 2. **Class selector (*.title*)** : Elements with `class="title"`
- 3. **ID selector (*#header*)** : Element with `id="header"`
- 4. **Group selector (*h1, p*)** : All `<h1>` and `<p>` elements
- 5. **Universal (***)** : Every element

ID (#ID)	CLASS (.CLASS)
Unique (one per page)	Reusable
High	Medium
Use only when needed	Recommended

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CSS Syntax



CSS Syntax — How Styling Works

The Basic Structure:

```
selector {  
  property: value;  
}
```

1. ***selector*** : Targets the HTML element you want to style
2. ***property*** : What you want to change (color, size, etc.)
3. ***value*** : The value you assign to the property

CSS Syntax — How Styling Works

Example :

```
p {  
  color: red;  
  font-size: 18px;  
}
```

Notes:

- *Always end each property-value pair with a semicolon ;*
- *The block of styles must be wrapped in {}*
- *You can apply multiple properties to the same selector*

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Text & Color Styling



Text & Color Styling

It allows us to control the appearance of text elements by adjusting their color, size, alignment, and style

1. ***color*** : Text color
2. ***background-color*** : Background color
3. ***font-size*** : Size of text
4. ***text-align*** : Alignment (left, center...)
5. ***font-family*** : which font is used to display the text
6. ***font-weight*** : Boldness of text
7. ***font-style*** : Italic
8. ***text-decoration*** : Underline, line-through etc.

Text & Color Styling

Notes & Resources:

01. Need Fancy Fonts :

Use Google Fonts to find and use free, modern fonts in your project.

Link : <https://fonts.google.com/>

02. Need Color Inspiration :

Check these awesome websites to explore and copy beautiful color palettes:

Link : coolors.co

Link : colorhunt.co

Link : paletton.com

Link : flatuicolors.com

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Box & Spacing



Box & Spacing — Control Layout Like a Pro

In CSS, every HTML element is treated as a box This model lets you control spacing, borders, and content.

1. ***border***: Line around the element
2. ***padding*** : Space ***inside*** the element (internal space)
3. ***margin*** : Space ***outside*** the element (external gap)
4. ***width/height*** : Controls the element's size

Real-Life Analogy: "The House Layout" :

1. ***Content*** : The actual house 🏠 (what's inside)
2. ***Padding*** : The walls & rooms inside the house 🏠 (space before touching the walls)
3. ***Border*** : The walls of the house or fence 🚧
4. ***Margin*** : The garden or yard outside the house 🌳

S01

Thank you for attending
this presentation!